

SDD REPORT AI-based Interview Practice and Feedback System

BS Information Technology

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1. Introduction

The AI-Based Interview Practice and Feedback System is a web-based application designed to help students and job seekers improve their interview preparation skills. The system provides an interactive environment where users can practice interview questions related to different job roles and receive automated feedback on their responses. By utilizing Natural Language Processing (NLP) techniques, the system evaluates user answers and generates performance scores and improvement suggestions. The application aims to enhance interview readiness, confidence, and communication skills through technology-driven learning.

1.1 Purpose of the Document

The purpose of this Software Design Document (SDD) is to describe the design and architecture of the AI-Based Interview Practice and Feedback System. This document serves as a blueprint for the development process by defining system components, database structure, module design, user interface design, and interactions between different parts of the system. It provides a clear understanding of how the system will be implemented and maintained.

1.2 Scope of the System

The AI-Based Interview Practice and Feedback System will provide users with a platform to practice interview questions based on selected job roles. The system will allow users to register, log in, select a job category, answer interview questions, and receive automated feedback and scoring. The application will use NLP techniques to analyze responses and provide meaningful suggestions for improvement. The scope includes web-based implementation, database management, user authentication, answer analysis, and feedback generation.

1.3 Definitions, Acronyms, and Abbreviations

1.4 References

The following references were used during the design and planning of the system:

Python Documentation – <https://docs.python.org>

Flask Documentation – <https://flask.palletsprojects.com>

NLTK Documentation – <https://www.nltk.org>

HTML, CSS, and JavaScript Documentation – <https://developer.mozilla.org>

GitHub Documentation – <https://docs.github.com>

1.5 Overview

This document presents the complete design of the AI-Based Interview Practice and Feedback System. It includes system architecture, database design, module design, user interface design, functional design, security considerations, testing strategy, deployment plan, and maintenance requirements. The document serves as a technical guide for the development and implementation of the proposed system.

2. System Overview

The AI-Based Interview Practice and Feedback System is a web-based application developed to assist students and job seekers in preparing for interviews. The system provides an interactive platform where users can practice interview questions related to specific job roles and receive automated feedback on their responses. By utilizing Natural Language Processing (NLP), the system evaluates user answers, identifies relevant keywords, and generates scores and suggestions for improvement.

The application aims to simulate a real interview environment, allowing users to improve their communication skills, technical knowledge, and confidence before attending actual interviews.

2.1 System Description

The system consists of a frontend interface, backend processing layer, NLP analysis module, and database. Users interact with the system through a web browser where they can register, log in, select a job role, answer interview questions, and receive feedback. The backend processes user requests, communicates with the database, and sends responses to the NLP module for analysis. The system architecture includes:

Frontend developed using HTML, CSS, and JavaScript

Backend developed using Python and Flask

NLP module implemented using NLTK

Database implemented using SQLite/MySQL

GitHub for version control and project management

The system is designed to be user-friendly, scalable, and accessible through modern web browsers.

2.2 System Objectives

The primary objectives of the AI-Based Interview Practice and Feedback System are:

To provide an interactive platform for interview preparation.

To allow users to practice interview questions based on different job roles.

To analyze interview responses using Natural Language Processing techniques.

To generate automated feedback and performance scores.

To improve users' communication and interview skills.

To increase confidence among students and job seekers before real interviews.

To provide a simple and accessible web-based solution for interview training.

2.3 User Characteristics

The system supports the following user categories:

Students

Primary users of the system.

Possess basic computer and internet skills.

Use the platform to prepare for internships and job interviews.

Job Seekers

Individuals actively searching for employment opportunities.

Use the system to improve interview performance and confidence.

Require practical interview preparation and feedback.

Administrator

Responsible for managing interview questions and system data.

Has moderate technical knowledge.

Monitors system activities and maintains database records.

2.4 Assumptions and Constraints

Assumptions

Users have access to a computer or mobile device with an internet connection.

Users possess basic knowledge of web browsing and online applications.

The database contains sufficient interview questions for different job roles.

Users provide honest and meaningful responses during interview practice sessions.

Constraints

The system is limited to a web-based platform and does not include a mobile application.

The NLP analysis is based on predefined techniques and keyword evaluation.

Real-time voice analysis and speech recognition are not included in the current version.

Internet connectivity is required to access system features.

The effectiveness of feedback depends on the quality and completeness of user responses.

3. System Architecture

The AI-Based Interview Practice and Feedback System follows a multi-layer architecture that separates the user interface, business logic, answer analysis, and data storage components. This architecture improves maintainability, scalability, and system performance while ensuring smooth communication between different modules.

The system is designed as a web-based application where users interact with the frontend interface, while the backend processes requests, manages data, and performs answer analysis through Natural Language Processing techniques.

3.1 Architectural Design

The system adopts a client-server architecture consisting of four major layers:

Presentation Layer (Frontend)

The presentation layer provides the user interface through which users interact with the system. It includes registration forms, login pages, interview question screens, and feedback pages.

Application Layer (Backend)

The application layer handles business logic, user authentication, question retrieval, answer processing, and communication with the database.

NLP Processing Layer

This layer is responsible for analyzing user answers using Natural Language Processing techniques. It extracts keywords, evaluates responses, and generates performance scores and feedback.

Data Layer

The data layer stores all system information, including user accounts, interview questions, submitted answers, and feedback records. The layered architecture ensures that each component operates independently while maintaining efficient communication between modules.

3.2 System Components

The system consists of the following main components:

User Interface Component

Provides web pages for:

User Registration

User Login

Job Role Selection

Interview Practice

Feedback Display

Authentication Component

Responsible for:

User registration

User login

Session management

Access control

Interview Management Component

Responsible for:

Managing interview questions

Loading questions according to selected job roles

Recording user responses

NLP Analysis Component

Responsible for:

Processing user answers

Extracting keywords

Evaluating answer quality

Calculating performance scores

Feedback Component

Responsible for:

Generating feedback

Displaying scores

Providing improvement suggestions

Database Component

Responsible for:

Storing user information

Managing interview questions

Saving user answers

Maintaining feedback records

3.3 Technology Stack

The system will be implemented using the following technologies:

These technologies are selected because they are lightweight, reliable, easy to learn, and suitable for developing intelligent web-based applications.

3.4 Architecture Diagram

The following diagram represents the overall system architecture:

The user interacts with the web interface, which sends requests to the Flask backend. The backend communicates with the NLP module for answer analysis and the database for data storage and retrieval. The generated feedback is then presented to the user.

4. Database Design

The database plays a vital role in the AI-Based Interview Practice and Feedback System by storing and managing all application data. It maintains information related to users, job roles, interview questions, user responses, and feedback scores. The database design ensures efficient data retrieval, consistency, and secure storage of information.

4.1 Database Architecture

The system uses a relational database architecture implemented using SQLite or MySQL. The database is connected to the Flask backend, which performs all Create, Read, Update, and Delete (CRUD) operations. The database consists of four primary entities:

Users

Job Roles

Questions

Answers

These entities are interconnected through primary and foreign key relationships to maintain data integrity and consistency.

Database Flow

User



Flask Backend



Database

■■■ Users

■■■ Job Roles

■■■ Questions

■■■ Answers

The backend communicates with the database whenever users register, log in, answer questions, or request feedback.

4.2 ER Diagram

The Entity Relationship (ER) Diagram represents the relationship between different database entities.

Relationships

One User can submit multiple Answers.

One Question can have multiple Answers.

One Job Role can contain multiple Questions.

Each Answer belongs to one User and one Question.

4.3 Tables and Relationships

Table 1: Users

Table 2: Job_Roles

Table 3: Questions

Table 4: Answers

4.4 Data Dictionary

Users Table

Job_Roles Table

Questions Table

Answers Table

The database structure is designed to support efficient storage, retrieval, and analysis of interview practice data while maintaining data consistency and scalability.

5. Module Design

The AI-Based Interview Practice and Feedback System is divided into multiple modules to ensure efficient functionality, maintainability, and scalability. Each module performs a specific task and collaborates with other modules to provide a complete interview preparation platform.

5.1 User Module

Purpose

The User Module manages all user-related activities within the system.

Responsibilities

User Registration

User Login

Profile Management

Job Role Selection

Interview Participation

Viewing Feedback and Scores

Inputs

User Name

Email Address

Password

Selected Job Role

Interview Answers

Outputs

User Account Creation

Interview Questions Display

Performance Scores

Feedback Reports

5.2 Admin Module

Purpose

The Admin Module provides administrative control over the system and manages all essential resources.

Responsibilities

Manage Users

Manage Interview Questions

Manage Job Categories

Monitor User Activities

Maintain System Data

Inputs

Question Details

Job Role Information

User Information

Output

Updated Question Database

Updated Job Categories

User Management Report

Benefits

Centralized management of system resource

Easy maintenance of interview content

Better control over application data

5.3 Authentication Module

Purpose

The Authentication Module ensures secure access to the system by validating user credentials.

Responsibilities

User Registration Verification

User Login Authentication

Session Management

Access Control

Inputs

Email Address

Password

Outputs

Login Success/Failure Message

User Session Creation

Security Features

Password Encryption

Input Validation

Unauthorized Access Prevention

5.4 Reporting Module

Purpose

The Reporting Module generates performance reports and feedback based on interview practice sessions.

Responsibilities

Generate Interview Scores

Display Performance Reports

Provide Improvement Suggestions

Track User Progress

Inputs

User Answer

NLP Analysis Results

Outputs

Interview Score

Feedback Report

Performance Summary

Benefits

Helps users identify strengths and weaknesses

Supports continuous improvement

Tracks interview preparation progress

5.5 Other Functional Modules

Interview Management Module

Purpose

Manages interview questions and interview sessions.

Responsibilities

Load Questions Based on Job Role

Display Interview Questions

Record User Responses

NLP Analysis Module

Purpose

Analyzes interview answers using Natural Language Processing techniques.

Responsibilities

Process User Responses

Extract Important Keywords

Compare Answers with Expected Responses

Calculate Score

Technologies Used

Python

NLTK

Feedback Generation Module

Purpose

Provides meaningful feedback based on answer evaluation.

Responsibilities

Generate Performance Score

Suggest Areas for Improvement

Display Results to User

Database Management Module

Purpose

Handles storage and retrieval of application data.

Responsibilities

Store User Information

Store Interview Questions

Store User Answers

Store Generated Scores and Feedback

Benefits

Secure Data Management

Fast Data Retrieval

Improved System Reliability

The modular architecture ensures that each component operates independently while maintaining smooth communication with other modules. This design improves maintainability, scalability, and future enhancement capabilities of the system.

6. User Interface Design The User Interface (UI) of the AI-Based Interview Practice and Feedback System is designed to be simple, user-friendly, and responsive. It allows users to easily navigate through the system and access interview practice features.

6.1 UI Overview The system provides an interactive interface for users to register, log in, select job roles, answer interview questions, and view feedback. Main Screens • Registration Page • Login Page • Dashboard • Interview Practice Page • Feedback Page • Admin Dashboard

6.2 Screen Designs Registration Page Allows new users to create an account by entering their personal information. Login Page Allows registered users to access the system using their credentials. Dashboard Displays available job roles and interview options. Interview Page Shows interview questions and allows users to submit answers. Feedback Page Displays scores, feedback, and improvement suggestions.

6.3 Navigation Structure Home ↓ Register/Login ↓ Dashboard ↓ Select Job Role ↓ Interview Questions ↓ Feedback & Score

6.4 User Experience Considerations • Easy-to-use interface • Responsive design • Fast navigation • Clear instructions • Consistent layout and design The UI is designed to provide a smooth and efficient experience for users preparing for interviews.

7. Functional Design The functional design describes the main functionalities of the AI-Based Interview Practice and Feedback System and how users interact with the system.

7.1 Functional Requirements The system shall provide the following functionalities: • User Registration • User Login • Job Role Selection • Interview Question Display • Answer Submission • NLP-Based Answer Analysis • Feedback Generation • User Progress Tracking • Admin Management of Questions and Users

7.2 Use Case Diagrams The Use Case Diagram illustrates the interaction between users and the system. Main Use Cases User • Register Account • Login • Select Job Role • Answer Questions • View Feedback Admin • Manage Users • Manage Questions • View System Data

7.3 Activity Diagrams The Activity Diagram shows the workflow of the interview practice process. Activity Flow

7.4 Sequence Diagrams The Sequence Diagram shows the communication between system components during an interview session. Sequence Flow User → Web Interface Web Interface → Backend Backend → Database Backend → NLP Module NLP Module → Backend Backend → User The system processes user responses, analyzes them using NLP, and returns feedback and scores.

8. Security Design The Security Design of the AI-Based Interview Practice and Feedback System ensures that user information and system data remain protected from unauthorized access, modification, and loss. Appropriate security measures are implemented to maintain confidentiality, integrity, and availability of data.

8.1 Authentication and Authorization The system uses authentication mechanisms to verify user identities before granting access. Features • User Registration and Login • Password Verification • Session Management • Role-Based Access (User and Admin) Only authorized users can access protected system features.

8.2 Data Security The system protects user data through secure storage and handling practices. Security Measures • Password Encryption • Input Validation • Secure Database Access • Protection against unauthorized data modification These measures help ensure the confidentiality and integrity of stored information.

8.3 Backup and Recovery Backup and recovery mechanisms help prevent data loss and support system restoration in case of failures. Features • Regular Database Backups • Data Recovery Procedures • Backup Storage Management These practices ensure that important system data can be restored when required.

8.4 Risk Management The system identifies potential risks and applies preventive measures to minimize their impact. Possible Risks • Unauthorized Access • Data Loss • System Failure • Incorrect User Input Mitigation Strategies • Strong Authentication • Regular Backups • Input Validation • Continuous System Monitoring These measures improve system reliability and reduce security-related risks.

9. Performance and Scalability The AI-Based Interview Practice and Feedback System is designed to provide efficient performance and support future growth. The system aims to deliver quick responses, reliable operations, and the ability to accommodate increasing numbers of users and data.

9.1 Performance Requirements The system should provide fast and efficient services to users. Requirements • Quick user authentication and login. • Fast retrieval of interview questions. • Efficient processing of user responses. • Timely generation of feedback and scores. • Smooth navigation between system pages. The system should maintain acceptable response times during

normal usage. 9.2 Scalability Considerations The system is designed to support future expansion and increasing user demand. Considerations • Ability to add more interview categories and job roles. • Support for a larger number of users. • Expansion of the question database. • Integration of advanced AI and NLP features in future versions. • Migration to larger database systems if required. The modular architecture allows the system to grow without significant redesign. 9.3 Reliability and Availability The system should provide stable and consistent service to users. Reliability Features • Accurate data storage and retrieval. • Consistent system behavior during operations. • Error handling and validation mechanisms. Availability Features • Web-based access from any location with internet connectivity. • Continuous availability during normal operating conditions. • Regular maintenance to ensure uninterrupted service. These measures help ensure that the system remains dependable and accessible for users.

10. Testing Strategy The testing strategy ensures that the AI-Based Interview Practice and Feedback System functions correctly, meets user requirements, and performs reliably. Different testing techniques will be applied to identify and resolve errors before deployment. 10.1 Unit Testing Unit Testing is performed on individual modules of the system to verify that each component works correctly. Modules Tested • User Registration Module • Login Module • Job Role Selection Module • Answer Submission Module • Feedback Generation Module Objective To ensure that each module performs its intended function without errors. 10.2 Integration Testing Integration Testing verifies that different modules communicate and work together correctly. Areas Tested • Frontend and Backend Integration • Backend and Database Integration • Backend and NLP Module Integration Objective To ensure smooth data flow between all system components. 10.3 System Testing System Testing evaluates the complete application as a whole. Areas Tested • User Registration and Login • Interview Question Display • Answer Analysis • Feedback Generation • Database Operations Objective To verify that the entire system functions according to the specified requirements. 10.4 User Acceptance Testing User Acceptance Testing (UAT) is conducted by potential users to ensure that the system meets their expectations and requirements. Participants • Students • Job Seekers • Project Supervisor Objective • Validate system usability. • Confirm that system requirements are satisfied. • Gather feedback for improvements. Successful completion of User Acceptance Testing indicates that the system is ready for deployment and practical use.

11. Deployment Design The deployment design describes how the AI-Based Interview Practice and Feedback System will be installed, configured, and made available to users. The system is designed as a web-based application that can be accessed through standard web browsers. 11.1 Deployment Architecture The system follows a client-server deployment architecture. Components • Client Device (Web Browser) • Web Server (Flask Application) • Database Server (SQLite/MySQL) • NLP Processing Module (NLTK) Deployment Flow User Browser ↓ Web Application (Flask) ↓ Database (SQLite/MySQL) ↓ NLP Module (NLTK) Users access the system through a web browser, while the backend processes requests and communicates with the database and NLP module. 11.2 Installation Requirements The following software and tools are required for installation and execution of the system: Software Requirements • Python 3.x • Flask Framework • NLTK Library • SQLite/MySQL Database • Visual Studio Code • GitHub (Version Control) Hardware Requirements • Processor: Intel Core i3 or higher • RAM: 4 GB or higher • Storage: 10 GB free disk space • Internet Connection 11.3 Hosting Environment The system can be deployed on a local machine or a cloud hosting platform. Supported Environments • Local Development Server • Shared Hosting • Cloud Hosting Services Browser Support • Google Chrome • Microsoft Edge • Mozilla Firefox The hosting environment should provide reliable access, adequate storage, and support for Python-based web applications to ensure smooth system operation. 12. Maintenance and Support Maintenance and support are essential to ensure the continuous operation, reliability, and improvement of the AI-Based Interview Practice and Feedback System. Regular maintenance activities help keep the system updated, secure, and efficient. 12.1 Maintenance Plan The system will be regularly monitored and maintained to ensure smooth operation. Maintenance Activities • Updating interview questions and job categories. • Fixing software bugs and errors. • Improving system performance. • Updating security features. • Monitoring database health and storage. Regular maintenance will help ensure that the system remains functional and up-to-date. 12.2 Future Enhancements The system can be enhanced in

future versions by adding new features and technologies. Possible Enhancements • Voice-based interview practice. • Speech recognition and analysis. • Advanced AI-powered feedback generation. • Mobile application development. • Real-time interview simulation. • Performance analytics and progress tracking. • Additional job categories and question sets. These enhancements will improve the overall effectiveness and usability of the system. 12.3 Support Requirements Technical support is required to ensure uninterrupted system operation and user satisfaction. Support Activities • Assisting users with technical issues. • Resolving system errors and bugs. • Managing software updates. • Monitoring system performance. • Maintaining database backups. Proper support ensures that users can effectively utilize the system and receive a reliable interview preparation experience.

13. Conclusion

The AI-Based Interview Practice and Feedback System is designed to help students and job seekers improve their interview preparation through an interactive and intelligent platform. The system allows users to practice interview questions related to different job roles and receive automated feedback based on their responses.

The proposed design incorporates a user-friendly interface, secure authentication mechanisms, efficient database management, and Natural Language Processing techniques for answer analysis. The modular architecture ensures scalability, maintainability, and future enhancements.

Overall, the system provides a practical and innovative solution for interview preparation by combining modern web technologies with AI-based feedback generation. It aims to enhance users' confidence, communication skills, and overall interview performance.

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